Dolan School of Business

DATA 3335 - Sports Analytics - Spring 2026

Professor Information

Name	Office Hours
Bianca (Gadže) Meyer	4-7 PM Monday Reserve via calendly or message via slack.

Slack: #data3335-spring26 ***Please use Slack as the main method of communication***
Google Classroom Meeting Code: ysc5ytsv (link)

Course Information

Meeting Days: Wednesday 11:00 AM - 1:30 PM || **Location:** DSBN 215
Credit Hours: 3 || **Prerequisites:** DATA 1101 || **Course Duration:** 01/20/2026 - 05/08/2026

This course explores how data is reshaping the world of sports. Students will learn the fundamentals of collecting, analyzing, and communicating insights from sports data through hands-on projects. This course emphasizes practical applications of analytics to real-world questions faced by teams, athletes, leagues, and front offices. By the end of the semester, students will have built a portfolio of projects showcasing skills in data wrangling, visualization, predictive modeling, and storytelling, all grounded in the context of sports performance, strategy, and business.

Course Materials

Technical Requirements

Assignments may require the use of computing software, in particular Microsoft Excel; make sure you have access to Microsoft prior to our first meeting. For more information on downloading Office 365 through Fairfield University, visit https://my.fairfield.edu/.

Learning Objective

Upon completion of this course, the students should be able to:

- Collect and clean sports datasets from public APIs, play-by-play logs, and tracking systems
- Apply statistical and machine learning methods to answer questions about player evaluation, team strategy, injury risk, fan engagement, or sports business
- Build compelling visualizations (shot charts, heat maps, etc.) tailored for various audiences
- Communicate findings through presentations, dashboards, and written reports that balance analytical rigor with accessibility
- Work collaboratively on a sports analytics project from idea to execution

Topic Plan | Course Schedule

I reserve the right to change following topics when deemed necessary / depending on interest.

	Date	Topic	Assignment	Assignment Due
1	Jan 21	Syllabus, Intro to Sports Analytics	• Onboarding Survey • HW 1	-
2	Jan 28	Sports Data Sources & Generative AI Tools	• HW 2 • Checkpoint 1	• Onboarding Survey • HW 1
3	Feb 4	Initial Exploratory Data Analysis	• Checkpoint 2 & 3	• HW 2 • Checkpoint 1
4	Feb 11	Cleaning Sports Data | From Data to Insights: Initial Insights & Correlation | Version Control	• Mini Project 1 • Checkpoint 4	• Checkpoint 2 & 3
5	Feb 18	Data Transformation | Relational Datasets | Time-Series Analysis	• Mini Project 2 • Checkpoint 5	• Mini Project 1 • Checkpoint 4
6	Feb 25	Feature Engineering | Multiple Regression | Text Data in Sports	• HW 3 • Checkpoint 6	• Mini Project 2 • Checkpoint 5
7	Mar 4	Multivariate Analysis & Probabilistic Models	**Enjoy the break :)**	• HW 3
-	Mar 11	😎☀️No Class - Spring Break ☀️😎		
8	Mar 18	Clustering & PCA: Finding Patterns	• HW 4 • Checkpoint 7	• Checkpoint 6
9	Mar 25	Classification & Statistical Testing	• Mini Project 3 • Checkpoint 8	• HW 4 • Checkpoint 7
10	Apr 1	Game Theory & Expected Value	• Mini Project 4 • Checkpoint 9 • Checkpoint 10	• Checkpoint 8 • Mini Project 3
-	Apr 8	No Class - University Operates on a Monday Schedule		
11	Apr 15	Simulation for Prediction	• Mini Project 5 • Checkpoint 11 • Checkpoint 12	• Mini Project 4 • Checkpoint 9 • Checkpoint 10
12	Apr 22	Final Project Workshop & Q&A	• Final	• Mini Project 5 • Checkpoint 11
13	Apr 29	Course Wrap-Up & Project Conclusions	**Final Project Presentations**	• Checkpoint 12 • Final Project Presentation

Grading Policies & Evaluation

Professionalism: (5% of Course Grade)

Given the large class enrollment and technical nature of the subject, it is critical that everyone acts in a professional manner. Students are expected to come to class prepared and participate to the best of their abilities. Cheating, free-riding, and other unprofessional behavior will not be tolerated.

Homeworks: (20% of Course Grade)

Students will have homework assignments within Google Classroom. When assigned, responses must be submitted prior the start of the upcoming class. Each homework assignment is graded of equal weight. Vacuous or empty responses or simple emojis count as zero, the same as no responses.

Mini Projects: (35% of Course Grade)

Throughout the course, you will complete mini projects designed to apply the concepts and techniques we've discussed in class to real-world sports analytics scenarios. Each mini project will focus on a different aspect of sports data analysis, such as statistical modeling, data visualization, or predictive analytics. These projects will allow you to demonstrate your understanding and creativity in solving practical problems in sports analytics.

Semester Project: (40% of Course Grade)

The semester project is the capstone of this course, your opportunity to apply what you've learned to a real-world sports analytics problem. Over the semester, you will frame a meaningful research question, work with authentic sports data, apply advanced analytical techniques, and communicate your findings in a professional format. The ultimate goal is for you to walk away with a portfolio-ready project you can showcase to future employers, graduate programs, or industry professionals. The project grade is divided into three key components.

Components of the Semester Project

Detailed formatting and submission guidelines will be posted in Google Classroom.

1. *Weekly Checkpoints (20%)*
    - To ensure consistent progress and high-quality work, the project is scaffolded into weekly checkpoints. These checkpoints walk you step-by-step through the research process. Each checkpoint builds directly into your final deliverable. Submitting work on time will keep you on track, allow for feedback, and prevent "last-minute crunching."
2. *Final Deliverable (10%)*
    - Your final submission will be a professional research paper that brings together all stages of your semester project into a cohesive and polished narrative. The paper should demonstrate both technical rigor and communication clarity, and must include:
        i. **Introduction & Motivation:** Why your research question matters in the context of sports analytics

ii. **Literature Review:** Connections to relevant prior work and how your project builds upon or differs from it
iii. **Methodology:** Description of your dataset, preprocessing steps, and analytical models used
iv. **Results:** Findings presented with clear, well-labeled visualizations and thoughtful interpretation
v. **Discussion & Conclusion:** Implications of your findings, limitations of your approach, and directions for future exploration

- The purpose of this paper is to create a **portfolio-ready artifact**, something you can showcase in interviews, share with prospective employers, or highlight as evidence of your applied skills.

3. *Final Presentation (5%)*

4. *Personal Reflection (5%)*

   - After completion, each student will submit an individual written reflection addressing:

     i. Skills you developed or improved
     ii. Challenges you encountered and how you addressed them
     iii. What you learned about working with messy sports data
     iv. How this project contributes to your growth as an analyst

   - The reflection ensures you not only demonstrate technical growth, but also articulate personal and professional development

Detailed criteria and guidelines will be provided for each component of the final project, including checkpoints, the presentation update, and the final deliverable. These criteria will outline expectations for content, analysis, and presentation to help you successfully meet the objectives of the project. Be sure to review the criteria, each week, as they are made available to ensure your work aligns with the course standards.

Grading Scale

Percentages and letter grades are provided for your convenience.

Percentage	Letter Grade	Percentage	Letter Grade
93 - 100	*A*	*77 - 79.9*	*C+*
90 - 92.9	*A-*	*73 - 76.9*	*C*
87 - 89.9	*B+*	*70 - 72.9*	*C-*
83 - 86.9	*B*	*60 - 69.9*	*D*
80 - 82.9	*B-*	*59.9 and below*	*F*

Other Course Policies

Laptops and Phones: You will need to be able to launch and use Microsoft Excel at the very least, so be sure you can do that on your computer. It is your responsibility to make sure that your laptop is up-to-date so that it runs the system smoothly. Other than that, if you want to use your laptop or phone or any other electronics or just doodle or daydream, that's all entirely up to you. Welcome to adulting.

Course Materials: All the teaching materials, including assignments and exams, are the intellectual properties of Fairfield University, distributing such materials without approval is strongly prohibited.

Late Submissions: No late submissions of anything.

Course Communications: Join the Slack at https://join.slack.com/t/fairfield-dolan-u/shared_invite/zt-3cpu74b8k-gY_7QFdoKnRKi6OwXREMfg and join the #data3335-spring26 public channel. Also join the Google Classroom https://classroom.google.com/c/ODM5MjgzMDg5OTU3?cjc=ysc5ytsv

Academic Honesty: Fairfield University's primary purpose is the pursuit of academic excellence. All members of the Fairfield University community share responsibility for establishing and maintaining appropriate standards of academic honesty and integrity. This is possible only in an atmosphere where discovery and communication of knowledge are marked by scrupulous, unqualified honesty. In the event of such dishonesty, professors are to award a grade of zero for the project, paper, or examination in question, and may record an ***F*** for the course itself.

Students with Disabilities: Fairfield University is committed to providing qualified students with disabilities an equal opportunity to access the benefits, rights and privileges of its services, programs and activities, in an accessible setting. Furthermore, in compliance with Section 504 of the Rehabilitation Act, the Americans with Disabilities Act, and Connecticut State laws, we provide reasonable accommodations to qualified students, to reduce the impact of disabilities on academic functioning or upon other major life activities.

Mental Health: Fairfield University provides mental health services to support the academic and personal success and well-being of students. Counseling & Psychological Services offers free, confidential psychological services to help students manage personal challenges that may interfere with your well-being. Fairfield University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of support, services are available.

Add / Drop Policy
Students are responsible for understanding the policies and procedures about add/drops, academic renewal, etc. found in the Fairfield University Catalog. You should be aware of the new deadlines and penalties for adding and dropping classes.

Other Course Policies (continued)

ADA Policy
Fairfield University is committed to achieving equal educational opportunities and providing students who have documented disabilities equal access to all University programs, services and activities. For this course to be accessible to all students, different accommodations or adjustments may need to be implemented. If you require accommodations for this course because of a disability, please contact the Office of Accessibility as early as possible this semester as they are the designated department on campus responsible for approving and coordinating reasonable accommodations for students with disabilities. Accessibility will help you understand your rights and responsibilities under the American with Disabilities Act and Section 504 of the Rehabilitation Act and will provide you further assistance with requesting and arranging accommodations for your courses. Once you receive your accommodations letter, please make an appointment to meet with me as soon as possible so that we can review your accommodations together and discuss how best to help you achieve equal access in this course this semester. Office of Accessibility, Library, 203-254-4081 (tel:2032544081), ooa@fairfield.edu

Religious Holidays
As indicated by Fairfield University's policy, any student missing classes or other assigned work due to religious holiday observances shall have the opportunity to make up that work during the term. It is the student's responsibility to notify the instructor in writing at least two weeks prior to any observed holiday.

Other
A note on sports betting: because it is such an integral part of sports analytics, we will cover some of the fundamentals of sports betting. However, please be aware of the risks and dangers of sports betting, which can include addiction, monetary loss, and many other possibilities including possible legal ramifications since sports betting is still not legal on all sports for all people in all regions. Nothing in this course is intended to encourage or promote sports betting, and, other than one lesson, nothing in this course will require you to do anything involving even learning about sports betting.